Investigation: IGB ELETRONICA S.A - EM RECUPERACAO JUDICIAL (CVM code 6815), fiscal year 2010, as it was on 2024-12-31

Statistical signal based on public financial statements. Not an accusation, not investment advice.

Company

Sector: Comunicação e Informática (origin: FCA); group: non-financial.
Trading codes (2023 FCA, version 2, filed on 2023-06-01): IGBR3 (Ações Ordinárias, Bolsa, since 1973-12-19).

Document used

2010 DFP, version 2, filed on 2011-04-27; origin: open data; type: consolidated; original scale: UNIDADE.
Versions known on the date: 1 (2011-04-27); 2 (2011-04-27).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=6725

Possible manipulation (engine v1, study 1)

Beneish M-score
- Value: component not computable: DEPI
- Alert by the study 1 rule: does not trigger alerts
- Rule: M > −1.78 (pre-registration, section 4)
- Percentile in sector and year: —
- Percentile in group and year: —
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: low

DSRI, days' sales in receivables index
- Value: 16.691 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: fewer than 10 companies with a computable value
- Percentile in group and year: 99 (n = 464)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

GMI, gross margin index
- Value: 2.552 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: fewer than 10 companies with a computable value
- Percentile in group and year: 98 (n = 474)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

AQI, asset quality index
- Value: 0.387 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: fewer than 10 companies with a computable value
- Percentile in group and year: 2 (n = 478)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

SGI, sales growth index
- Value: 0.068 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: fewer than 10 companies with a computable value

Percentile in group and year: 0 (n = 475)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DEPI, depreciation index
Value: missing account: depreciation, cash flow lines in section 6.01.01
M-score component; it has no rule of its own
Percentile in sector and year: —
Percentile in group and year: —
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: low

SGAI, SG&A expenses index
Value: 11.917 (index)
M-score component; it has no rule of its own
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 100 (n = 472)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

TATA, total accruals to total assets
Value: 0.140 (index)
M-score component; it has no rule of its own
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 94 (n = 501)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

LVGI, leverage index
Value: 0.747 (index)
M-score component; it has no rule of its own
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 8 (n = 488)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DPO jump
Value: 0.0 days
Alert by the study 1 rule: no
Rule: DPO change in the year's top decile among non-financial companies (≥ 90th percentile of computable values on the date: 19.1 days)
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 49 (n = 446)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium
Alternative explanations: Bargaining power with suppliers and a change in purchase mix (decision of 2026-11-07)

Implied interest rate minus average CDI
Value: 0.80 p.p.
Alert by the study 1 rule: no
Rule: implied interest rate above the year's average CDI rate + 10 p.p. (pre-registration, section 4)
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 25 (n = 442)
The company's own history: fewer than 5 previous fiscal years with a value

Confidence in the data: medium
Alternative explanations: Expensive debt, exchange differences and lease interest inside finance costs (decision of 2026-11-07)

Accruals to average total assets
Value: 0.156 (index)
Alert by the study 1 rule: does not trigger alerts
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 94 (n = 494)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

Financial risk (context only)

Altman Z'' (financial risk, context only)
Value: −13.527 (index)
Components: X1 = −0.205; X2 = −3.672; X3 = 0.089; X4 = −0.770 (indices)
Percentile in sector and year: 11 (n = 617); percentile in the non-financial universe: the sector has fewer than 10 companies with a value
The company's own history: fewer than 5 previous fiscal years with a value
- No cut-off zone: the available source does not define zones for Z''
- Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

Financial quality (context only)

Piotroski F (financial quality, context only)
Value: F = 2/9 (5 computable)
Signals:
- Return on assets (ROA): 0
- Operating cash flow on assets (CFO): 0
- Change in ROA (ΔROA): —
- Cash above earnings (ACCRUAL): 0
- Change in leverage (ΔLEVER): —
- Change in current ratio (ΔLIQUID): 1
- No equity issuance (EQ_OFFER): —
- Change in gross margin (ΔMARGIN): 1
- Change in asset turnover (ΔTURN): —
- No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement

Changes

Highlighted in the following year's comparative: 7 (threshold of R$0.0 million, 1% of total assets of the version used). Values in R$ million:
BPP 2.02 Passivo Não Circulante: 0.6 in the version used (v2, filed on 2011-04-27); 550.5 in the next DFP's comparative (v2, filed on 2013-11-26); difference of 549.9. Different value in the following year's comparative (restatement or reclassification).
BPP 2.03 Patrimônio Líquido Consolidado: −0.5 in the version used (v2, filed on 2011-04-27); −451.7 in the next DFP's comparative (v2, filed on 2013-11-26); difference of −451.2. Different value in the following year's comparative (restatement or reclassification).
BPA 1 Ativo Total: 0.1 in the version used (v2, filed on 2011-04-27); 135.2 in the next DFP's comparative (v2, filed on 2013-11-26); difference of 135.0. Different value in the following year's comparative (restatement or reclassification).

BPP 2.01 Passivo Circulante: 0.0 in the version used (v2, filed on 2011-04-27); 36.4 in the next DFP's comparative (v2, filed on 2013-11-26); difference of 36.4. Different value in the following year's comparative (restatement or reclassification).
DFC (indirect method) 6.01 Caixa Líquido Atividades Operacionais: −0.0 in the version used (v2, filed on 2011-04-27); −32.3 in the next DFP's comparative (v2, filed on 2013-11-26); difference of −32.3. Different value in the following year's comparative (restatement or reclassification).
DRE 3.11 Lucro/Prejuízo Consolidado do Período: −0.0 in the version used (v2, filed on 2011-04-27); −13.4 in the next DFP's comparative (v2, filed on 2013-11-26); difference of −13.4. Different value in the following year's comparative (restatement or reclassification).
DRE 3.01 Receita de Venda de Bens e/ou Serviços: 0.0 in the version used (v2, filed on 2011-04-27); 0.3 in the next DFP's comparative (v2, filed on 2013-11-26); difference of 0.3. Different value in the following year's comparative (restatement or reclassification).

Highlighted: differences in totals and key lines (total assets, current liabilities, non-current liabilities, equity, net revenue, net income and operating cash flow) above 1% of total assets of the version used. The other differences (reclassifications between lines and smaller differences) stay in the full list.
Full list: 63 accounts above the R$1,000 tolerance (group a, totals and key lines: 7; group b, reclassifications: 56), in the annex at the end.

What was known on the date

- FCA 2010, v1: 2010-11-01
- DFP 2010, v1: 2011-04-27
- DFP 2010, v2: 2011-04-27
- DFP 2011, v1: 2012-04-13
- DFP 2011, v2: 2013-11-26

DFP and FCA filed with the CVM by the reference date; ITR and material facts come in later phases.

Data quality

53 applicable checks; 0 failed.

Auditor's report and auditor (factual section; no alert)

no opinion excerpt was found in the text
Auditor in the 2021 FCA: Grant Thornton Auditores Independentes Ltda.
Auditor in the 2022 FCA: Grant Thornton Auditores Independentes Ltda.
Auditor in the 2023 FCA: Grant Thornton Auditores Independentes Ltda.
Auditor changes (one line per change, canonical name by the firm's CVM code):

- 2013: BC Control Auditores Independentes -> individual auditor (name omitted). The outgoing firm audited 1 consecutive fiscal years (2010 to 2010), by the start and end dates in the FCA.
- 2014: individual auditor (name omitted) -> Verdus Auditores independentes. The outgoing firm audited 2 consecutive fiscal years (2012 to 2013), by the start and end dates in the FCA.
- 2018: Verdus Auditores independentes -> Conatus Auditores Independentes S.S. The outgoing firm audited 5 consecutive fiscal years (2014 to 2018), by the start and end dates in the FCA. The change coincides with the end of the 5-fiscal-year term and is compatible with the rotation rule (art. 31 of CVM Instruction No. 308/1999, in force when the new firm started).
- 2020: Conatus Auditores Independentes S.S -> Grant Thornton Auditores Independentes Ltda. The outgoing firm audited 2 consecutive fiscal years (2019 to 2020), by the start and end dates in the FCA.

Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

How the engine was tested

In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

Limits

- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

Limitations

- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 36 DFP lines, each with document, version, filing date, account and location.

Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a E). Generated at 2026-11-10T19:37:38+00:00.

Today's information (outside the reference date)

Registry status: CANCELADA. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.

Annex: full list of differences in the following year's comparative (R$ million)

(a) BPP 2.02 Passivo Não Circulante: 0.6 in the version used, 550.5 in the comparative (difference of 549.9)
(a) BPP 2.03 Patrimônio Líquido Consolidado: −0.5 in the version used, −451.7 in the comparative (difference of −451.2)
(a) BPA 1 Ativo Total: 0.1 in the version used, 135.2 in the comparative (difference of 135.0)
(a) BPP 2.01 Passivo Circulante: 0.0 in the version used, 36.4 in the comparative (difference of 36.4)
(a) DFC (indirect method) 6.01 Caixa Líquido Atividades Operacionais: −0.0 in the version used, −32.3 in the comparative (difference of −32.3)
(a) DRE 3.11 Lucro/Prejuízo Consolidado do Período: −0.0 in the version used, −13.4 in the comparative (difference of −13.4)
(a) DRE 3.01 Receita de Venda de Bens e/ou Serviços: 0.0 in the version used, 0.3 in the comparative (difference of 0.3)
(b) BPP 2.03.05 Lucros/Prejuízos Acumulados: −0.5 in the version used, −496.4 in the comparative (difference of −495.9)
(b) BPP 2.02.01 Empréstimos e Financiamentos: 0.4 in the version used, 395.0 in the comparative (difference of 394.6)
(b) BPP 2.02.01.01 Empréstimos e Financiamentos: 0.4 in the version used, 395.0 in the comparative (difference of 394.6)
(b) BPP 2.02.01.01.01 Em Moeda Nacional: 0.4 in the version used, 395.0 in the comparative (difference of 394.6)
(b) BPP 2 Passivo Total: 0.1 in the version used, 135.2 in the comparative (difference of 135.0)

(b) BPA 1.02 Ativo Não Circulante: 0.1 in the version used, 126.4 in the comparative (difference of 126.3)
(b) BPA 1.02.03 Imobilizado: 0.1 in the version used, 111.8 in the comparative (difference of 111.6)
(b) BPA 1.02.03.01 Imobilizado em Operação: 0.1 in the version used, 111.8 in the comparative (difference of 111.6)
(b) BPP 2.02.02 Outras Obrigações: 0.1 in the version used, 84.6 in the comparative (difference of 84.5)
(b) BPP 2.02.02.01 Passivos com Partes Relacionadas: 0.1 in the version used, 84.6 in the comparative (difference of 84.5)
(b) BPP 2.02.02.01.01 Débitos com Coligadas: 0.1 in the version used, 84.6 in the comparative (difference of 84.5)
(b) BPP 2.02.04 Provisões: 0.1 in the version used, 70.9 in the comparative (difference of 70.8)
(b) BPP 2.02.04.01 Provisões Fiscais Previdenciárias Trabalhistas e Cíveis: 0.1 in the version used, 70.9 in the comparative (difference of 70.8)
(b) DRE 3.06.02 Despesas Financeiras: −0.0 in the version used, −42.6 in the comparative (difference of −42.6)
(b) BPP 2.03.01 Capital Social Realizado: 0.0 in the version used, 41.0 in the comparative (difference of 41.0)
(b) DRE 3.06 Resultado Financeiro: −0.0 in the version used, −25.4 in the comparative (difference of −25.3)
(b) DFC (indirect method) 6.02 Caixa Líquido Atividades de Investimento: 0.0 in the version used, 23.0 in the comparative (difference of 23.0)
(b) DFC (indirect method) 6.01.02 Variações nos Ativos e Passivos: −0.0 in the version used, −22.4 in the comparative (difference of −22.4)
(b) BPP 2.01.03 Obrigações Fiscais: 0.0 in the version used, 18.8 in the comparative (difference of 18.8)
(b) BPP 2.01.03.01 Obrigações Fiscais Federais: 0.0 in the version used, 18.8 in the comparative (difference of 18.8)
(b) BPP 2.01.03.01.01 Imposto de Renda e Contribuição Social a Pagar: 0.0 in the version used, 18.8 in the comparative (difference of 18.8)
(b) DRE 3.06.01 Receitas Financeiras: 0.0 in the version used, 17.2 in the comparative (difference of 17.2)
(b) DRE 3.04.06 Resultado de Equivalência Patrimonial: 0.0 in the version used, 16.4 in the comparative (difference of 16.4)
(b) BPA 1.02.01 Ativo Realizável a Longo Prazo: 0.0 in the version used, 14.7 in the comparative (difference of 14.7)
(b) BPA 1.02.01.01 Aplicações Financeiras Avaliadas a Valor Justo: 0.0 in the version used, 14.7 in the comparative (difference of 14.7)
(b) DRE 3.07 Resultado Antes dos Tributos sobre o Lucro: −0.0 in the version used, −13.4 in the comparative (difference of −13.4)
(b) DRE 3.09 Resultado Líquido das Operações Continuadas: −0.0 in the version used, −13.4 in the comparative (difference of −13.4)
(b) DRE 3.11.01 Atribuído a Sócios da Empresa Controladora: −0.0 in the version used, −13.4 in the comparative (difference of −13.4)
(b) BPP 2.01.01 Obrigações Sociais e Trabalhistas: 0.0 in the version used, 13.0 in the comparative (difference of 13.0)
(b) BPP 2.01.01.02 Obrigações Trabalhistas: 0.0 in the version used, 13.0 in the comparative (difference of 13.0)
(b) DRE 3.04 Despesas/Receitas Operacionais: 0.0 in the version used, 12.5 in the comparative (difference of 12.5)
(b) DRE 3.05 Resultado Antes do Resultado Financeiro e dos Tributos: 0.0 in the version used, 12.0 in the comparative (difference of 12.0)
(b) DFC (indirect method) 6.01.01 Caixa Gerado nas Operações: −0.0 in the version used, −9.8 in the comparative (difference of −9.8)
(b) DRE 3.04.01 Despesas com Vendas: −0.0 in the version used, −9.4 in the comparative

(difference of −9.4)
(b) DFC (indirect method) 6.03 Caixa Líquido Atividades de Financiamento: 0.0 in the version used, 9.2 in the comparative (difference of 9.2)
(b) BPA 1.01 Ativo Circulante: 0.0 in the version used, 8.7 in the comparative (difference of 8.7)
(b) DRE 3.04.05 Outras Despesas Operacionais: 0.0 in the version used, 7.9 in the comparative (difference of 7.9)
(b) BPP 2.01.04 Empréstimos e Financiamentos: 0.0 in the version used, 4.2 in the comparative (difference of 4.2)
(b) BPP 2.01.04.01 Empréstimos e Financiamentos: 0.0 in the version used, 4.2 in the comparative (difference of 4.2)
(b) BPP 2.01.04.01.01 Em Moeda Nacional: 0.0 in the version used, 4.2 in the comparative (difference of 4.2)
(b) BPA 1.01.08 Outros Ativos Circulantes: 0.0 in the version used, 4.0 in the comparative (difference of 4.0)
(b) BPA 1.01.08.03 Outros: 0.0 in the version used, 4.0 in the comparative (difference of 4.0)
(b) BPP 2.03.02 Reservas de Capital: 0.0 in the version used, 3.7 in the comparative (difference of 3.7)
(b) DRE 3.04.02 Despesas Gerais e Administrativas: −0.0 in the version used, −2.4 in the comparative (difference of −2.4)
(b) BPA 1.01.04 Estoques: 0.0 in the version used, 2.3 in the comparative (difference of 2.3)
(b) BPA 1.01.03 Contas a Receber: 0.0 in the version used, 2.1 in the comparative (difference of 2.1)
(b) BPA 1.01.03.01 Clientes: 0.0 in the version used, 1.7 in the comparative (difference of 1.7)
(b) DRE 3.02 Custo dos Bens e/ou Serviços Vendidos: −0.0 in the version used, −0.8 in the comparative (difference of −0.8)
(b) DRE 3.03 Resultado Bruto: −0.0 in the version used, −0.5 in the comparative (difference of −0.5)
(b) BPA 1.01.03.02 Outras Contas a Receber: 0.0 in the version used, 0.4 in the comparative (difference of 0.4)
(b) BPP 2.01.05 Outras Obrigações: 0.0 in the version used, 0.4 in the comparative (difference of 0.4)
(b) BPP 2.01.05.01 Passivos com Partes Relacionadas: 0.0 in the version used, 0.4 in the comparative (difference of 0.4)
(b) DFC (indirect method) 6.05.01 Saldo Inicial de Caixa e Equivalentes: 0.0 in the version used, 0.4 in the comparative (difference of 0.4)
(b) BPA 1.01.01 Caixa e Equivalentes de Caixa: 0.0 in the version used, 0.4 in the comparative (difference of 0.4)
(b) DFC (indirect method) 6.05.02 Saldo Final de Caixa e Equivalentes: 0.0 in the version used, 0.4 in the comparative (difference of 0.4)
(b) DFC (indirect method) 6.05 Aumento (Redução) de Caixa e Equivalentes: −0.0 in the version used, −0.0 in the comparative (difference of −0.0)